TCS CORPORATE WEBSITE

Columns Container Block — Developer Guide

Role / audience	Development — Developers, technical leads, block owners, AI coding agents
Version	1.0
Status	Draft for review
Owner	Technical Lead / Block Owner (proposed)
Last validated	2026-11-08
Documents	1
Classification	INTERNAL — classification to be confirmed

Generated from the Markdown sources of docs/handover (branch migrated-page-content-edam); do not edit this PDF.

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Columns Container Block — Developer Guide

Attribute	Value
Document ID	DEV-B-columns-container
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	How the Columns Container block is built: files, model, structure, behaviour, variations, tests and limitations
Scope	blocks/columns-container/ and its components columns-container
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	2026-11-08 (model, files, imports from the code; reference pages from the live site)
Source files	blocks/columns-container/README.md, blocks/columns-container/_columns-container.json, blocks/columns-container/columns-container.css, blocks/columns-container/columns-container.js
Related documents	Variations, Test coverage, Authoring guide
Related AEM paths	/content/tcs-corp/design-block-master/en/block-library/columns-container, /content/tcs-corp/design-block-master/en/block-library/columns-container/alignment-gap-and-breakpoints, /content/tcs-corp/design-block-master/en/block-library/columns-container/columns-of-blocks

1. Summary

Attribute	Value
Block	Columns Container
Folder	`blocks/columns-container`
Type	Container (section)
Category	container
Status	Active
Purpose	A section laid out in columns of default content and blocks
Variations (summary)	Columns per row, gap, vertical alignment, side by side from, BG Color
Children	column-break (Column Break), the components of its filter
Code size	194 lines of JS and CSS

2. Overview

Business purpose. Put content, including blocks such as Cards, Teasers, Lists or forms, side by side, with a responsive layout chosen by the author, without a purpose-built block.

User experience. Columns of equal width, stacked one under the other below the chosen breakpoint and side by side above it, in rows of Columns per row; optionally on a grey, dark or dark green background.

Templates / page types. Any page; no template adds it. **Data source:** authored content; placeholders for editor hints.

Related blocks. Column Break (starts each column), Tabs and Tab (inside a tab, Column Breaks build this block in the panel), Columns (the built-in block it replaces), Teaser (shares the BG Color values).

When to use. Two to four pieces of content side by side, especially when a column holds a block; contact details next to a form; a teaser next to a list; three cards of different kinds in a row.

When not to use. A grid of similar cards (use Cards or List, which have per-breakpoint column settings); an image with text and calls to action in the design-system style (use Teaser or Hero); tabular data (use Table); columns of unequal widths (the columns are always of equal width).

Realistic use cases. An "Our offices" section with an address Text in one column and an embedded map in the other; a dark green band with a heading and text in one column and a Teaser in the other; six highlights in two rows of three.

3. Source files

File	Link
`README.md`	GitHub
`_columns-container.json`	GitHub
`columns-container.css`	GitHub
`columns-container.js`	GitHub

4. Components, models and filters

Component	Title	Resource type	Template name	Model	Children allowed
`columns-container`	Columns Container	`core/franklin/components/section/v1/section`	Columns Container	`columns-container`	column-break, text, image, button-block, title, hero, teaser, image-collage, quote, media-contacts, testimonial, anchor-list, accordion, tabs-block, table, table-list, list, separator, embed, video, modal, form-variations, filter-list, conversation-canvas, cards, fragment

5. Fields

The Additional tab fields (anchor, hide, padding, margin, animation) are common to the blocks and described in the content modeling guide.

5.1 Columns Container (`columns-container`)

Tab	Label	Property	Type	Required in model	Default	Options	Visibility
Content	Style	`style`	multiselect	No	`columns-container-section`	2 options	—
Content	BG Color	`containerBgColor`	select	No	`""`	5 options	—
Content	Width Type	`sectionWidth`	select	No	`full-width`	2 options	—

Tab	Label	Property	Type	Required in model	Default	Options	Visibility
Content	Background Image	`backgroundImage`	reference	No	—	—	—
Content	Image alt text	`backgroundImageAlt`	text	No	—	—	—
Content	Anchor name	`anchorName`	text	No	—	—	—
Layout	Columns per row	`columnsPerRow`	select	No	`auto`	5 options	—
Layout	Gap	`columnsGap`	select	No	`medium`	4 options	—
Layout	Vertical alignment	`columnsAlign`	select	No	`top`	4 options	—
Layout	Side by side from	`columnsFrom`	select	No	`desktop`	3 options	—
Additional	Anchor ID	`containerAnchor`	text	No	—	—	—
Additional	Hide block	`containerHide`	boolean	No	`false`	—	—
Additional	Animation Type	`containerAnimation`	select	No	`""`	11 options	—
Additional	Motion Speed	`containerAnimationSpeed`	select	No	`""`	3 options	Conditional
Additional	Animation Trigger	`containerAnimationTrigger`	select	No	`""`	2 options	Conditional
Additional	Padding left	`containerPaddingLeft`	select	No	`""`	15 options	—
Additional	Padding right	`containerPaddingRight`	select	No	`""`	15 options	—
Additional	Padding top	`containerPaddingTop`	select	No	`""`	15 options	—
Additional	Padding bottom	`containerPaddingBottom`	select	No	`""`	15 options	—
Additional	Margin left	`containerMarginLeft`	select	No	`""`	15 options	—
Additional	Margin right	`containerMarginRight`	select	No	`""`	15 options	—
Additional	Margin top	`containerMarginTop`	select	No	`""`	15 options	—
Additional	Margin bottom	`containerMarginBottom`	select	No	`""`	15 options	—

"Required in model" is the model's `required` flag; whether a field is required in practice is stated in the authoring guide where it is known.

6. Dependencies

Import
`../../utils/placeholders.js`
`../column-break/column-break.js`

7. For Developers

```
blocks/columns-container/
├── README.md
├── _columns-container.json
├── columns-container.css
└── columns-container.js
```

Decorator/Utility	Purpose	Execution or dependency point
utils/placeholders.js	getPlaceholders, loadPlaceholders, fillPlaceholder (UI text from the locale's placeholders, with English fallbacks)	columns-container.js: { getPlaceholders }
blocks/column-break/column-break.js	—	columns-container.js: { readColumn }

Responsibilities.

- scripts/scripts.js buildColumnsSections: a section with the class columns-container-section (added to any section with the container's Layout data attributes) gets a synthetic div.columns-container; .column blocks are renamed column-break; the Layout values become classes (per-row-{v}, gap-{v}, align-{v}, from-{v}); applySectionBlockOptions adds the container* classes (BG Color, anchor, hideblock, motion, spacing); wrapSectionContent decorates the block while empty and moves the section's wrappers into it (blocks among them are decorated).
- columns-container.js: each wrapper holding a :scope > .column-break starts a column (div.columns-container-column; plain, not a block, so Universal Editor updates of the components in it find their own block); the Column Break's wrapper stays first in its column (the label). readColumn caches the Column Break's name; its hideblock and spacing classes are copied to the column; its Anchor ID (the id the block options gave it, else the anchor- class) moves to the column. An empty column gets p.columns-container-hint (columnEmpty); no column at all: the hint columnsContainerEmpty. The block gets columns-container-{n} (perRow: per-row-1…4, else the number of columns, at most MAX_COLUMNS 4).
- columns-container.css: scoped to .block.columns-container (AEM also marks the section of every Columns block columns-container, which turned those sections into grids before the block audit of 19 October 2026); a one-column grid, the column template from the breakpoint class; BG Color, gap and alignment classes; blocks fill their column; editor outlines and hints under html.adobe-ue-edit; hidden columns.

Rendered DOM.

```
<div class="columns-container block per-row-2 gap-large align-middle from-tablet bg-dark-green pt-1000
columns-container-2">
  <div class="columns-container-column">
    <div class="column-break-wrapper"><div class="column-break block" data-analytics="off">…</div></div>
    <div class="teaser-wrapper">…</div>
  </div>
  <div class="columns-container-column hideblock">…</div>
</div>
```

Decoration flow. The section's blocks load in order: the container first (it groups the wrappers), then the Column Breaks (their label uses the column index) and the blocks in the columns. In a tab, blocks/tabs/tabs.js creates the block with only per-row-N and loads it with loadBlock.

Exports. default decorate(block) only; it imports readColumn from blocks/column-break/column-break.js.

Error handling. No network; missing names or options fall back to defaults.

8. Extending the Block

1. **A Layout option:** add the field and options to `_columns-container.json`, run `npm run build:json` and `npm run lint`; read the data attribute in `buildColumnsSections` (add it to the selector that marks container sections) and add the class rule to `columns-container.css` with the `.block.columns-container` scope and the project breakpoints (600 / 900 / 1200px).
2. **Additional tab fields:** sections don't get `classes_*` delivered: use `container*` names; `applySectionBlockOptions` turns their values into classes (`containerhide` → `hideblock`).
3. **Delivery rules:** no field names ending in Text, Title, Type, Alt or MimeType for values to read; no sample values in definition templates; template values are stored on new containers (`columnsGap` medium, `columnsAlign` top, `columnsFrom` desktop, `columnsPerRow` auto).
4. **Motion:** the columns are the motion targets (`CONTENT_TARGETS['columns-container']` in `decorators/motion-decorator.js`); keep `.columns-container-column` or update the decorator.
5. **Theme:** new blocks inside should use the text tokens so they follow the dark backgrounds.
6. **Tests:** `test/motion/motion.test.mjs` checks the container fields (no `classes_*` on the section model); `test/models/block-names.test.mjs` checks that template names render as existing block folders (the "Column" → `column` defect). Check the block library pages listed below.
7. **Backward compatibility:** keep reading `.column` items and ignoring `columnsLayout`.
8. Update this README and run `npm run build:docs`; `npm run build:modules` after changing imports.

9. Responsive Behavior

Viewport	Layout	Image behavior	Interaction behavior
Below 600px	One column: the columns stacked in order, separated by the gap	Images and blocks at the column (full) width	As the components
600px to 899px	Side by side with Side by side from Tablet; stacked otherwise	Fill their column	As the components
900px to 1199px	Side by side with Tablet or Desktop (default); stacked with Large desktop	Fill their column	As the components
1200px and wider	Side by side with every setting, `Columns per row` per row	Fill their column	As the components

10. Accessibility

- **Semantics:** the columns are plain `div`s; the reading and focus order is the authored order at every width. Headings come from the components.
- **Keyboard and focus:** the block handles no keys; focus moves through the components in order.
- **Contrast:** BG Color Dark and Dark green apply the theme contract (light text, link and focus tokens; underlined links in running text; inverted buttons).
- **Background image:** alt text from Image alt text, empty for decorative images.
- **Reduced motion:** the Animation Type doesn't play for visitors who prefer reduced motion, or in the editor.
- **Hidden columns:** Hide block on a Column Break removes the column from the page (`display: none`).
- **Reflow:** columns stack below the breakpoint; `minmax(0, 1fr)` and `min-width: 0` keep wide content inside its column.
- **Editor hints:** shown only in edit mode (placeholders `columnEmpty`, `columnsContainerEmpty`, `columnLabel`).

Author responsibilities: a logical order of the columns (it is also the mobile order); headings in sequence; alt text for meaningful images; enough padding on coloured backgrounds.

Developer responsibilities: keep the content order; keep the theme tokens for text in new blocks; keep the hints in placeholders and hidden outside edit mode.

11. Data Layer and Analytics

- **Component:** the synthetic block is added by `decorators/data-layer-decorator.js` as `tcs-corp/components/columns-container` (id: its Anchor ID, else `columns-container-{n}`; `dc:title`: its first heading). Inside a tab the id is `{panel id}-columns`.
- **Children:** the blocks in the columns are components whose `parentId` is the container. The columns are not components; Column Breaks are `data-analytics="off"`.
- **Events:** the block pushes no events itself; clicks, media, submits in its columns are measured centrally and attributed to the block they're in (default content: to the container).
- **Path:** Adobe Client Data Layer → Adobe Data Collection (Tags) → AEP Web SDK → Edge Network → Analytics / CJA is the intended path, but Tags and the Web SDK aren't configured yet (blocked: measurement plan). No consent mechanism is implemented.

Trigger	Data-layer event	Required properties	Analytics behavior
Page load	component state	`@type` `tcs-corp/components/columns-container`, `parentId` (page), `dc:title`	Inventory
Click on a link or button in a column	`cmp:click`	component (the block in the column, else the container), `xdm:linkURL`, `xdm:linkType`, `breakpoint`	Link / button clicks

12. Performance Considerations

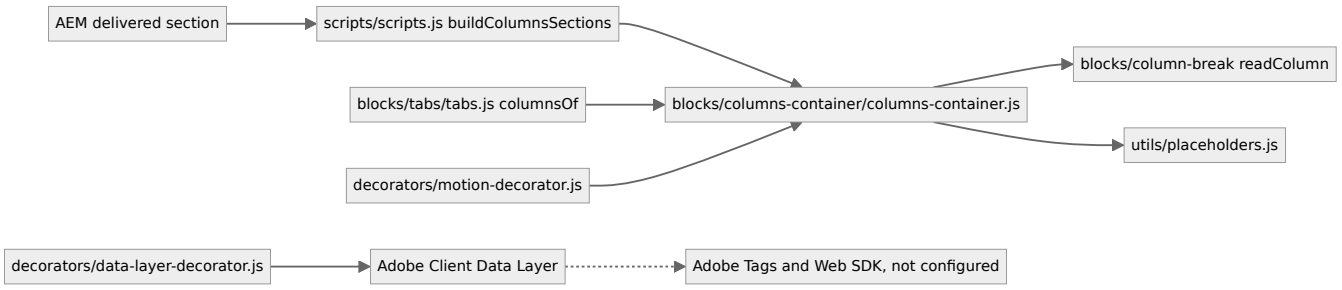
- Synchronous regrouping of existing DOM; one placeholders lookup (cached per locale); no network requests of its own, no listeners or observers.
- Loaded with its section (eager if it is the first section, else lazy); the block's CSS and JS are loaded once per page.
- The section stays hidden until its blocks are loaded, so the regrouping causes no visible layout shift; the column layout is pure CSS.
- The blocks in the columns load as in a section (sequentially in content order); many heavy blocks in one container load one after another.

13. Security and Privacy

No external content, query strings or user data; the editor hints are set with `textContent`. The components in the columns have their own handling.

14. For Architects

- **Role:** the supported way to put blocks side by side: Edge Delivery allows one level of nesting (blocks and default content in sections), so columns of blocks are a section container whose content is wrapped in a synthetic block, like the Tabs Container and Gen Carousel.
- **Content model:** section model `columns-container` (Layout and `container*` fields delivered as data attributes; Content fields as the Section's); item marker `column-break` (key-value block).
- **Rendering:** client-side regrouping of the delivered section; no data source.
- **Dependencies:** `blocks/column-break` (`readColumn`), `utils/placeholders.js`, `scripts/scripts.js`, `decorators/motion-decorator.js`, `decorators/data-layer-decorator.js`, `styles/styles.css`.
- **Data layer / Web SDK / Target:** ACDL component only; Tags / Web SDK not configured; no Target. **Consent / identity:** none.
- **Caching / security:** no requests of its own. **Scalability:** grid layout; no limit beyond the content.



15. Troubleshooting

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Content shown as a plain section	Not a Columns Container (Style removed and no Layout values)	Inspect the section for `columns-container-section`	Re-select Style Columns Container	Columns appear
Everything in one column	No Column Breaks	Check the content tree	Add a Column Break per column	Columns appear
An unexpected first column	Content before the first Column Break	Check the tree order	Move it after a Column Break	Expected columns
Columns stacked on tablet	Side by side from Desktop (default)	Check the Layout tab; inspect for `from-*`	Choose Tablet (600px)	Side by side at 600px
No space inside the colour	BG Color adds no padding	Check the Additional tab	Set Padding	Space inside
Padding / Hide / Animation without effect	Fields not delivered (content authored before the `container*` fields)	Inspect the section's `data-container*` attributes	Re-enter the values in the Additional tab	Classes on the block
Column Break option change not visible in the editor	Applied when the section is rendered again (R-C-7)	Reload	Reload	Applied

Evidence to collect (no secrets): page URL, viewport width, the section's classes and `data-*` attributes, the block's classes, console errors.

16. Known limitations of the block

Limitation	Impact	Workaround	Planned correction
Columns are always of equal width (Column widths removed in handover 2.26)	No 1/3 + 2/3 layouts	Use blocks with their own layouts (Teaser image position)	Not planned
Column Break option changes apply after the section is rendered again (re-audit R-C-7, deferred)	No immediate feedback in the editor	Reload the editor	Not recorded
At most 4 columns per row	Wider rows not possible	More rows	Not planned
Inside a tab only Columns per row is authorable	Gap, alignment and breakpoint fixed (Medium, Top, Desktop)	Use a Columns Container section	Not planned
A section can't be placed in a column	No Tabs Container, Gen Carousel or Columns Container in a column	Use the Tabs block in a column	Not planned

17. Change history of the block

Version	Change	Source	Date
Not recorded	README generated from verified implementation and approved handover documentation	Repository review	Not recorded
Handover 2.22	Columns Container introduced: columns of default content and blocks, each started by a Column item; Layout and Additional tab	docs/handover revision history	19 October 2026
Handover 2.23	Columns per row (Automatic, 1–4); rows follow from the number of Column items	docs/handover revision history	19 October 2026
Handover 2.24	The Column item is Column Break; columns are plain wrappers; content authored with "Column" still works (F-99)	docs/handover revision history	19 October 2026
Handover 2.25	The Section's Style, Background Image, Image alt text and Anchor name in the Content tab; a section with the Layout values stays a container	docs/handover revision history	19 October 2026
Handover 2.26	Column widths removed; columns are of equal width	docs/handover revision history	19 October 2026
Handover 2.27	Block audit: the CSS no longer turns sections with a Columns block into grids	docs/handover revision history	19 October 2026
Handover 2.28	Built-in Columns no longer offered; Column Breaks in tabs build a Columns Container	docs/handover revision history	20 October 2026
Handover 2.30	Additional tab fields are `container*` (AEM doesn't deliver a section's `classes_*`)	docs/handover revision history	20 October 2026
Handover 2.32	BG Color (`containerBgColor`) and Gap None	docs/handover revision history	20 October 2026

18. Reference pages

Kind	Page	Live URL
Block library	/design-block-master/en/block-library/columns-container	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/columns-container
Block library	/design-block-master/en/block-library/columns-container/alignment-gap-and-breakpoints	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/columns-container/alignment-gap-and-breakpoints
Block library	/design-block-master/en/block-library/columns-container/columns-of-blocks	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/columns-container/columns-of-blocks
Block library	/design-block-master/en/block-library/columns-container/columns-per-row	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/columns-container/columns-per-row
Curated design page	/design-block-master/en/curated-design-pages/industry-landing-page	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/curated-design-pages/industry-landing-page
Curated design page	/design-block-master/en/curated-design-pages/services	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/curated-design-pages/services
Curated design page	/design-block-master/en/curated-design-pages/story	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/curated-design-pages/story

Kind	Page	Live URL
Curated design page	/design-block-master/en/curated-design-pages/what-we-do	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/curated-design-pages/what-we-do

19. Coverage of the required topics

Topic	Where
Purpose	Overview
Source files	generated: Files
Model, definition, filters, children	generated: Components
Dependencies	generated: Dependencies
Authored input / rendered output / DOM transformation	For Developers
Configuration fields, defaults, required fields	generated: Fields
Variations	COLUMNS-CONTAINER-VARIATIONS.md
Responsive behaviour	Responsive Behavior
Accessibility and keyboard	Accessibility
Analytics / data layer	Data Layer and Analytics
Performance	Performance Considerations
Security	Security and Privacy
Extending	Extending the Block
Known limitations	Known Limitations
Troubleshooting	Troubleshooting
Tests	COLUMNS-CONTAINER-TEST-COVERAGE.md
SEO behaviour	Not recorded
Caching considerations	Not recorded (Edge Delivery default caching; no block-specific rule)
Ownership	Technical Lead / Block Owner (proposed)

20. Screenshots

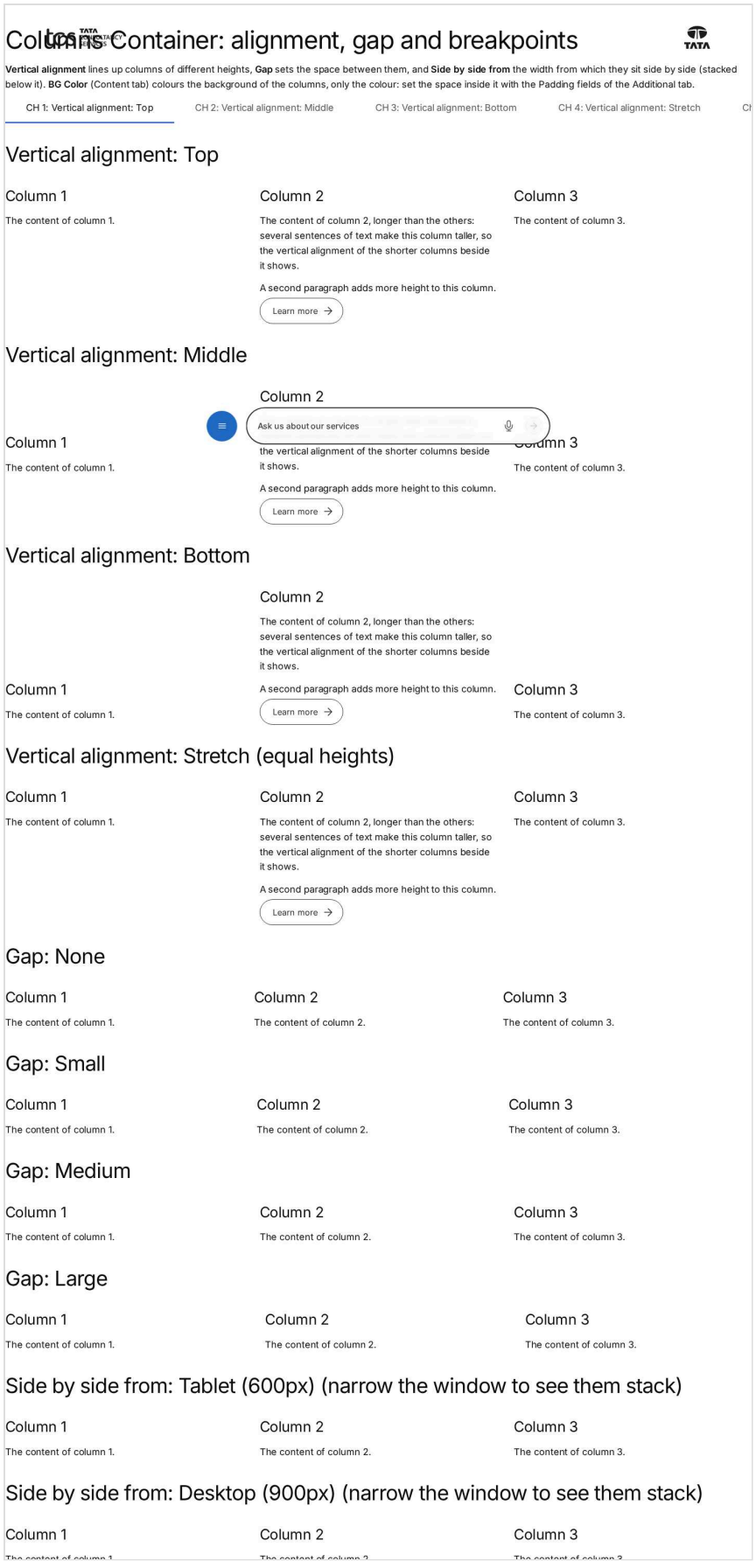


Figure: Columns Container block library page — desktop 1440 × 900 (/design-block-master/en/block-library/columns-container/alignment-gap-and-breakpoints, captured 2026-11-08).

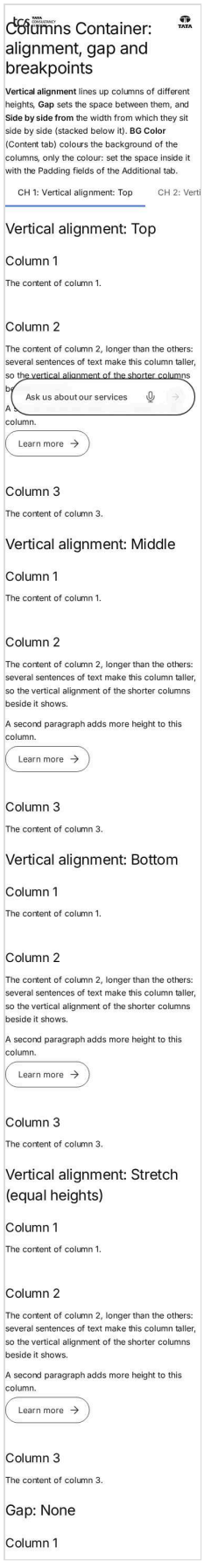


Figure: Columns Container block library page — mobile 375 × 812 (/design-block-master/en/block-library/columns-container/alignment-gap-and-breakpoints, captured 2026-11-08).

21. Known limitations

- Universal Editor dialogs were not captured: no author access in this program (see the access blockers).

22. Open items

- None.

23. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages